

The analytic amplitude adapter

Claude, with Daniel Murfet

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Abstract

An amplitude given on the box by a double power series with a radius margin, $\Phi(u, v, s) = \sum c_{ij}(s) u^i v^j$ with $|c_{ij}(s)| \rho^{i+j} \leq H(s)$ and $b < \rho$, is packaged into the data of the $d = 2$ cutoff theorem (Astra #6, route D): the jointly continuous amplitude `anaAmp` (coordinates clamped to $[0, b]$ so that the series converges everywhere), the compatible faces `anaFaceU`, `anaFaceV` of unit 113, the face remainders with envelope $H(s) \rho^{-i-M} / (1 - b/\rho)$, and the mixed remainder bound $H(s) \rho^{-M_1-M_2} / (1 - b/\rho)^2 \cdot u^{M_1} v^{M_2}$. Continuity in s comes from a locally uniform bound (`continuous_tsum_of_locally_bounded`), which needs only continuity of the coefficient functions and of H . Zero sorry/axiom.

1 The things formalised

```
theorem continuous_tsum_of_locally_bounded {X : Type*} [PseudoMetricSpace X]
  [ProperSpace X] {f : Type*} (f :  $\rightarrow X \rightarrow$ ) (hf :  $\mathbf{n}, \text{Continuous (f n)}$ ) (u :  $\rightarrow$ )
  (hu : Summable u)
  (hu0 :  $\mathbf{n}, 0 \leq u \mathbf{n}$ ) (G :  $X \rightarrow$ ) (hG : Continuous G)
  (hbound :  $\mathbf{n} \mathbf{x}, |f \mathbf{n} \mathbf{x}| \leq G \mathbf{x} * u \mathbf{n}$ ) :
  Continuous fun x => '  $\mathbf{n}, f \mathbf{n} \mathbf{x}$ 
theorem summable_geom_prod ( : ) (h0 : 0 < ) (h1 : < 1) :
  Summable fun ij :  $\times \Rightarrow \wedge ij.1 * \wedge ij.2$ 
noncomputable def clampB (b x : ) : := max 0 (min b x)
-- + continuity, bounds, identity on Icc
noncomputable def anaAmp (c :  $\times \rightarrow \rightarrow$ ) (b u v s : ) : :=
  dblSum (fun ij => c ij s) (clampB b u) (clampB b v)
noncomputable def anaFaceU (c :  $\times \rightarrow \rightarrow$ ) (b : ) (i : ) (v s : ) : :=
  faceU (fun ij => c ij s) i (clampB b v)
noncomputable def anaFaceV ... := faceV (fun ij => c ij s) j (clampB b u)
def swapC (c :  $\times \rightarrow \rightarrow$ ) :  $\times \rightarrow \rightarrow := fun ij s => c (ij.2, ij.1) s$ 
theorem anaAmp_continuous ... : Continuous fun x :  $\times \times \Rightarrow \text{anaAmp c b x.1 x.2.1 x.2.2}$ 
theorem anaFaceU_continuous / anaFaceV_continuous ... :
  Continuous (Function.uncurry (anaFaceU c b i))
theorem anaFaceU_rem ... (hv : v Ioc (0 : ) b) :
  |anaFaceU c b i v s -  $\mathbf{m} \text{ Finset.range } M, c (i, m) s * v ^ m|$ 
  H s * (  $\wedge i$ )  $^1$  * (  $\wedge M$ )  $^1$  / (1 - b / ) * v ^ M
theorem anaFaceV_rem ... -- via swapC
theorem anaAmp_rectRem_le ... (hu : u Icc (0 : ) b) (hv : v Icc (0 : ) b) :
  |rectRem (anaAmp c b) (anaFaceU c b) (anaFaceV c b) (fun i j s => c (i, j) s)|
  M M u v s|
  H s * (  $\wedge (M + M)$ )  $^1$  / (1 - b / ) ^ 2 * (u ^ M * v ^ M)
theorem anaCoeff_abs_le ... : |c ij s| H s * (  $\wedge (ij.1 + ij.2)$ )  $^1$ 
```

2 Mathematics

The series $\sum_{ij} c_{ij}(s)u^i v^j$ converges absolutely on $[0, b]^2$ with the s -dependent bound $H(s)\theta^{i+j}$, $\theta = b/\rho$; a global uniform bound is unavailable (the envelope grows with s), but on every closed ball H is bounded, and `continuousOn_tsum` on the ball plus `ContinuousOn.continuousAt` gives continuity at every point. Clamping the coordinates to $[0, b]$ makes the amplitude and faces globally defined continuous functions agreeing with the series on the box, which is all the cutoff theorem uses. The remainder bounds are unit 113's with $1 - v/\rho \geq 1 - b/\rho$ on $(0, b]$, and the v -face statements follow from the u -face ones for the swapped coefficient array.

3 Paper-facing meaning

With this adapter, the $d = 2$ equal-exponent cutoff expansion needs no compatibility assumption on face jets and no symmetry of mixed partial derivatives: analytic data with a radius margin, the paper's standing hypothesis, produce compatible faces by construction. The next unit discharges the `FaceMoments` through `faceMoments_of_envelope` (all coefficients are bounded by $C_0 \rho^{-i-j}(1+s)^D e^{\beta s L}$ when H is) and states the analytic $d = 2$ cutoff theorem; the weighted- ℓ^1 coefficient closure for $\eta e^{\beta s \xi}$ then supplies the series hypothesis from those of ξ, η .

4 Lean notes

- `continuous_tsum` needs a global bound; for envelopes growing in a parameter, prove continuity at each point from `continuousOn_tsum` on `Metric.closedBall x 1` (compact in a `ProperSpace`) and `IsCompact.exists_bound_of_continuousOn`.
- Clamping with `max 0 (min b x)` is simpler than `Set.projIcc` (no subtype coercions); `clampB_of_mem` restores the identity on the box.
- `field_simp` cannot cancel $1 - b/\rho$ without the normalised fact $\rho - b \neq 0$; a targeted `div_mul_eq_mul_div`, `div_mul_cancel` _ hne is more robust.
- `gcongr` side goals of the form $0 < v \wedge M$ need $0 < v$ as a bare hypothesis in context (`have hv0 : 0 < v := hv.1`).
- Symmetric statements for the two faces: define the swapped array once (`swapC`) and transport the bound with `add_comm`.